Comparing PCA, t-SNE and UMAP for Regression on the Diabetes Dataset

Benchmarking three dimensionality-reduction methods on their ability to preserve a linear target signal in the diabetes dataset.

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Overview

High-dimensional medical data is difficult to visualise and interpret directly. This project applies three dimensionality-reduction techniques — Principal Component Analysis (PCA), t-Distributed Stochastic Neighbor Embedding (t-SNE), and Uniform Manifold Approximation and Projection (UMAP) — to the diabetes dataset (442 patients, 10 baseline variables, one continuous disease-progression target). Each method compresses the inputs to one, two, and three dimensions, and linear regression is then fitted on those coordinates so that the mean squared error can serve as an objective measure of how much predictive signal each reduction preserves.

Goals

Determine which dimensionality-reduction method best preserves the linear relationship with a continuous target, and characterise how each method's main tuning parameter changes both the visual layout and the regression error.

Objectives

- Scale all variables to the range [0, 1] with MinMaxScaler and characterise the dataset through variance and correlation analysis.
- Apply Lasso regression on the full ten-variable set as a reference baseline, sweeping the penalty parameter alpha across 0, 1, 10, 100, 500, and 1000.
- Run PCA, t-SNE, and UMAP and study how perplexity (t-SNE) and n_neighbors (UMAP) affect the resulting scatter plots.
- Fit linear regression on one, two, and three reduced dimensions for each method and record MSE.
- Compile a comparison table of all three methods at three dimensions and draw conclusions about the suitability of each for regression tasks.

Approach

The notebook follows a single structured pipeline: data loading and scaling, exploratory correlation analysis, a Lasso baseline, then one section per reduction method. For each method the main hyperparameter is swept visually before regression is evaluated at increasing dimensionality. All code is self-contained in a single notebook with one task per cell and clear headings, making the progression easy to follow and reproduce.

Outcomes

- BMI and S5 are the two inputs most correlated with the target (r = +0.59 and +0.57); Lasso achieves its lowest error (MSE = 2859.7) at alpha = 0 with all ten variables.
- PCA's first two components explain approximately 69 percent of variance; linear regression on PC1 and PC2 gives MSE = 3833.5, a clear improvement over the mean-prediction baseline of 5929.9.
- t-SNE and UMAP produce near-baseline errors in two dimensions (MSE around 5917 and 4696 respectively) because their axes do not preserve global linear structure.
- t-SNE in three dimensions achieves the best three-dimensional result (MSE = 3450.1, R-squared = 0.42), narrowly beating PCA (MSE = 3804.5, R-squared = 0.36).
- The study confirms that PCA is the safer choice when linear predictability matters, while t-SNE and UMAP are better suited to visual cluster exploration.